Corpus Analysis of COVID-19 Related Loneliness on Twitter

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Abstract. Loneliness is a complex mental health issue involving feelings of isolation, disconnectedness, and a lack of purpose in life. The coronavirus disease-19 (COVID-19) pandemic and lockdowns significantly increased social isolation, contributing to a global rise in loneliness on social online platforms. However, research on mental health expressions in Arabic on social media is limited, exacerbating the lack of understanding of loneliness and mental health issues in these communities. Stigma and societal resistance in the Middle East hinder individuals from seeking mental health support, particularly noticeable amidst the challenges of the pandemic. Thus, this study aimed to conduct a corpus analysis of an open-source dataset to gain insights into the origin of how loneliness was expressed in Arabic on Twitter during the initial months of COVID-19 regulations and explored temporal trends to enhance comprehension of the prevalence of loneliness and mental health struggles during the initial months of the pandemic.

Keywords: Loneliness · COVID-19 · Corpus Analysis · Thematic Analysis · Twitter · Arabic

1 Introduction

Loneliness, a psychological state devoid of purpose, encompasses diverse dimensions, including the subjective perception of isolation and the experience of negative emotions intertwined with scarce social connections. Robert Stuart Weiss posits loneliness as a natural and universal phenomenon capable of affecting individuals irrespective of demographic characteristics [1]. The advent of the coronavirus-19 (COVID-19) pandemic dramatically altered the global landscape, introducing widespread lockdowns that precipitated increased social isolation. The surge in loneliness during the COVID-19 pandemic has become a poignant societal concern, with individuals grappling with isolation, social distancing measures, and disruptions to traditional social interactions [2]. Social media has emerged as a crucial platform for individuals to navigate and alleviate feelings of loneliness. While social media platforms like Facebook, Twitter, and Instagram can encourage excessive use and negative interactions, they can also serve as valuable outlets for expression [3]. The multilingual nature of social media platforms has

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enabled diverse voices to express the effects of isolation on mental wellbeing. Twitter is a popular choice for text analysis due to its short length of expression (140 characters) and worldwide use of Twitter for expressing people's opinions on different issues [4]. Examining specific aspects of loneliness expression on Twitter provides valuable insights into how individuals articulate and communicate their feelings in the digital realm. Burke and Kraut [5] have shed light on the patterns and nuances of loneliness expression on Twitter. Individuals often share personal reflections, emotional states, and experiences related to isolation and solitude through tweets [6]. The COVID-19 pandemic has heightened emotional struggles among younger individuals, as social isolation and loneliness cause depression and anxiety symptoms [7].

This study aimed to demonstrate how corpus linguistics tools could be used to find expressions of mental health struggles and indications of the origin of loneliness due to COVID-19 within the corpus. It also aims to show how AntConc could be used to analyze an Arabic dataset to conduct a thematic analysis and analyze temporal trends to increase our understanding of the expression of mental health struggles and loneliness.

2 Related Work

Recent advancements in Arabic Natural Language Processing (NLP) and social media analysis offer critical insights for investigating nuanced expressions of loneliness and emotional states. Abbes et al. [19] developed a dialectal Arabic irony corpus extracted from Twitter, a resource that could illuminate the subtleties of loneliness expression in Arabic social media. Additionally, Ahmed et al. [20] conducted a review of freely available Arabic corpora, essential for identifying data sources to analyze loneliness expressions. Furthermore, Alam et al. [21] explored propaganda detection in Arabic social media content, offering methodological perspectives beneficial for analyzing mental health-related expressions.

The impact of the COVID-19 pandemic on mental health, accentuated through social media discourse, is well-documented. Alam et al. [22] provided an overview of the COVID-19 infodemic, highlighting how misinformation influences public mental health perceptions, a crucial context for loneliness studies. Zaghouani [23] introduced a corpus for detecting youth depression in social media, directly relevant for understanding online loneliness expressions.

Methodologies for data annotation and corpus creation are pivotal for this study. Zaghouani and Awad [24] described the creation of an Arabic punctuated corpus, pertinent for processing the text in this research. Obeid et al. [25] introduced a manual Arabic diacritization system, Mandiac, crucial for nuanced text analysis. Additionally, Zaghouani et al. [26] discussed dialectal variation handling in Arabic text annotation, vital for social media analysis.

Hawwari et al. [27] discussed building a lexical semantic resource for Arabic morphological patterns, aiding in semantic analysis. Lastly, Zaghouani and Charfi [28] development of a multi-dialect Twitter corpus provides insights into demographic variations in social media discourse.These studies collectively provide a robust framework for understanding and analyzing expressions of loneliness and mental health in Arabic social media content, particularly during the challenging times of the COVID-19 pandemic.

3 Methodology

3.1 Dataset Collection

To conduct the corpus analysis, we used the open-source dataset ArCov-19, as described previously [8]. ArCOV-19, an extensive Arabic COVID-19 Twitter dataset, chronicled one year from January 27, 2020, to January 31, 2021, comprising a substantial repository of approximately 2.7 million tweets. We sampled the dataset for the months of March, April, May, and June of 2020, as this timeframe was the months of the COVID-19 lockdown [9–12]. Our dataset began on March 11, when COVID-19 was considered a pandemic by the World Health Organization (WHO) [13]. We collected a total of 889,306 tweets.

3.2 Corpus Analysis Techniques

We identified the dataset from March to June of 2020 as the target corpus and loaded the corpus into AntConc for each month, along with the reference corpus. AntConc conducts corpus linguistics, or the analysis of electronic texts, to provide commonalities in language. We used the Almaany Arabic Dictionary and the Cambridge Dictionary English to Arabic feature to create a list of words associated with loneliness in Arabic [14, 15]. We searched for words in English and selected the English to Arabic option on both sites. Upon searching for the words lonely, loner, and alone, we were provided with their Arabic equivalents. We looked at words associated with loneliness and searched and included the words that included results, such as lonely, loner, alone, and secluded. The synonyms of loneliness through both dictionaries resulted in the words provided in Table 1.

Table 1. Variations of loneliness-related words, as per Almaany and Cambridge dictionaries.

Word	Dictionary meaning
لوحدي	Variation of alone (Almaany)
وحيدا	Variation of lonely and loner (Almaany)
وحيدا	Variation of lonely and loner (Cambridge Dictionary)
لوحيدة	Variation of lonely (Cambridge dictionary)
واحد	Variation of lonely (Almaany)
الوحيد	Variation of lonely and alone (Almaany)
منعزل	Variation of lonely and secluded (Almaany)
معزول	Variation of lonely and secluded (Almaany)
لوحدي	Variation of lonely (Almaany)
متوحد	Variation of lonely (Almaany)
الوحيدة	Variation of lonely and alone (Almaany)

We used our list of words from Table 1 and ran an advanced and basic search. For the advanced search, we used the concordance tab to search for tweets and used the keyword in context (KWIC) feature. For a basic search, we found collocates and ran a keyword analysis to analyze the keyness of the words.

3.3 Concordance Search

Upon loading the words in the advanced search option, we were directed to the Concordance tab. This feature provided us with the frequency of each word, which would assist us in finding relevant tweets. We listed the frequency of each word, which consisted of a combined frequency of 18,299. The prevalence of the selected loneliness-related words was identified within the target corpus.

Additionally, we will discuss some examples of concordance search result for each word directly associated with loneliness, along with a translation of each tweet. After loading the list of words from Table 1, we used AntConc's advanced search feature to run a KWIC search and advanced search and loaded words to get the number of hits (17367 hits). We searched each word separately, demonstrating that sometimes the words were in the context of loneliness, while other times, the context had nothing to do with loneliness.

The term "لوحدي" ("Alone") offers a profound insight into the various dimensions of solitude during the COVID-19 pandemic. In the first instance, " كورونا طلعت اسوء مافينا. ناس بتتعمد تاذي غيرها ماشيه بمبدأ لو مت مش هاموت لوحدي. وناس بتبعد عن اقرب الناس ليها لمجرد اصابتهم" we see the pandemic revealing humanity's darker sides, where some individuals, driven by a sense of shared despair, choose to harm others, embodying the sentiment of not dying alone. In contrast, others isolate themselves from their loved ones merely due to the fear of transmission, illustrating a different facet of solitude rooted in protection and care. Transitioning to the term "وحيدا" ("Lonely and loner"), the example " كورونا علمتنا درساً في غاية الاهميه عندما تبقى وحيداً ستنجو وعندما تتبع القطيع ستموت" speaks to the broader societal lesson imparted by the pandemic: the survival advantage of remaining solitary over following the collective. This echoes the previous sentiment, where individual actions—whether self-serving or protective—underscore the nuanced implications of solitude in crisis times. Exploring "وحيد" ("Lonely and loner") further, we encounter a poignant narrative: " الملياردير البرتغالي أنطونيو فييرا صاحب بنك سانتاندير بالبرتغال توفي بسبب عدوى #كورونا الأربعاء الماضي ابنته كتبت في إحدى وسائل التواصل الاجتماعي: نحن #عائلة_ثرية لكن والدي توفي وحيدا ومختنقا يبحث عن شيء مجاني هو الهواء والمال بقي في المنزل." This account shows that the pandemic's isolating effect spares no one, regardless of their social or economic status, emphasizing the universal experience of solitude in suffering. Continuing with the exploration of solitude during the pandemic, the term "الوحيد" ("Lonely or loner") provides a personal perspective: "انا الوحيد اللي مو مهتم في مرض ذا ولا ادري وش الوضع." This individual feels uniquely detached from the pervasive concern about the disease, illustrating a solitary stance not just in physical terms but also in their emotional and informational disconnect from the collective experience.

The concept of "منعزل" ("Lonely or secluded") is depicted in the scenario " خلصنا من فلم #الحظر_الشامل وداخلين في مسلسل #الحظر_الجزئي ولا نزال نعاني من اكشن ودرامة ال

#فيروس_كورونا_المستجد في العالم ، وبعض الناس قلبها طرارة ومكافآت وكأنه بقوقعة بروحة منعزل عن البشر." Here, the narrative extends to the societal level, where the pandemic has created a continuum of isolation experiences, from comprehensive lockdowns to partial ones, affecting individuals' sense of seclusion in varying degrees.

The term "معزول" ("Lonely or secluded") further extends the conversation to a broader context with "# الشروق| حرب طبقية وعالم معزول.. سيناريوهات للمستقبل بعد انتهاء أزمة كورونا." This reflects on the aftermath of the pandemic, suggesting a future where isolation might not just be a temporary condition but a defining characteristic of societal structure, hinting at deeper, possibly enduring, divides.

In the expression "متوحد" ("Lonely"), the narrative becomes more personal and somewhat positive: " عادي قولوا عليا متوحدة بس منجد مرة مبسوطة في ذا الوضع.. ما ابغى الحياة ترجع طبيعية، هي كذا مرة تمام، بعيد عن البشر بعيد عن الالتزامات الاجتماعية." This individual finds contentment in their isolation, valuing the distance from societal norms and obligations, which offers a counter-narrative to the commonly perceived negativity associated with being alone.

Lastly, "الوحيدة" ("Lonely and alone") provides a stark depiction of isolation: " انا الوحيدة اللي في حجر صحي حق كورونا حقيقي مامعاي ولا حد بين اربع حيط في السرير اليوم كلو كورونا زاتو حا تعفيني." The individual's experience of quarantine embodies the extreme end of solitude, confined without any human contact, emphasizing the intense personal challenges posed by the pandemic.

Together, these examples span a spectrum of solitude experiences during the pandemic, from individual to societal levels, highlighting the complex and varied impacts of isolation on different individuals and communities.

3.4 Collocate Search

We ran a separate search for the collocates of each word through a basic search. كورونا (corona) was the most frequent collocate for the words لوحدي (alone), وحيد (lonely), وحيدة (lonely and loner), (lonely) واحد, (lonely) الوحيدة, and (lonely and alone) الوحيد. The collocate for the words وحيدا (lonely or loner) and منعزل (lonely or secluded) was the word في (of). For the word معزول (lonely or secluded), the collocate was the word بعد (after). For the word متوحد (lonely), across the four lockdown months, the most frequent collocate was the word البيت (the house). While the collocates for each searched word, along with frequency, we looked at the words الوحيد (lonely, masculine) and الوحيدة (alone or lonely, feminine) and their top collocate كورونا (corona). The results for الوحيد (alone or lonely, masculine) and الوحيدة (alone or lonely, feminine), along with the top collocate كورونا (corona) of both words.

3.5 Keyword Analysis

We ran a keyword analysis and listed the top 20 keywords in Table 2 according to keyness. Of the 20 keywords, many of the words were closely related and associated with COVID-19, such as كورونا (corona), فيروس (virus), كوفيد (COVID), بفيروس (with the virus), and الكورونا (The corona). A separate cluster of words closely related to public health included the words الصحة (health), حالة (case), المستجد (update), إصابة (infected), حالات(cases), and وفاة (death). The last word on the list is the word بالبيت (at home), commonly used with the twelfth keyword خليك (stay) as a hashtag #خليك - بالبيت (#stayathome) to express staying at home or informing others to stay at home, due to lockdown, isolation, or to not infect others. The top keyword was ك رونا (corona).

3.6 Theme Identification

We analyzed prevailing thematic elements within the target corpus. The analytical process involved an analysis of 8,000 tweets, with a particular analysis of 100 tweets per month for each keyword. We selected 10 random tweets from each page per keyword to accumulate 100 tweets for each keyword for each month of the preliminary four lockdown months. The concordance feature was employed to view multiple hits and provide an in-depth review of the tweets identified by the keywords. This synergy of actions was enhanced, maximizing our ability to extract meaningful insights from the data. Our analysis discerned prominent themes, such as despair in the face of uncertainty, grief for the loss of life, hope through faith, opinions on COVID-19, opinions on isolation, COVID-19 news, and fear of illness. These themes emerged through our meticulous examination of tweets within the corpus. Following categorizing 8,000 tweets based on the identified themes, we observed overarching patterns. The themes ranked as follows: opinions on COVID-19, opinion of isolation, hope through faith, grief for the loss of life, despair in the face of uncertainty, and fear of illness. The theme of COVID-19 news dominated 80% of the keywords, with exceptions for الله (God), البيت(the house), خليك (stay), and كلنا (everyone). The primary themes for the latter three key words were Hope through faith and opinions on COVID-19. The distribution of themes within the corpus according to the 7 themes is presented in Fig. 1.

3.7 Temporal Trends

We discerned a gradual upswing in the articulation of loneliness over the ensuing weeks. The identification of tweets associated with loneliness markedly intensified in May and June 2020. A noteworthy semantic distinction emerged in our examination, as the utilization of the term كورونا(corona) in tweets conveyed a more comfortable sentiment compared to the more serious tone invoked by the term كوفيد ١٩(COVID-19) associated with news-related tweets. This observation was exemplified by contrasting tweets from June 25, 2020, where “COVID-19” was featured in a news context, while “corona” was humorously employed.

Table 2. Keyword analysis of target corpus according to Keyness.

Rank	Keyword	English	Keyness
1.	كورونا	Corona	2484.103
2.	فيروس	Virus	864.639
3.	كوفيد	Covid	241.580
4.	الصحة	Health	232.589
5.	حالة	Case	222.321
6.	الله	God	204.463
7.	المستجد	Update	160.517
8.	الكويت	Kuwait	145.347
9.	إصابة	Infected	141.837
10.	الجديد	New	127.127
11.	بفيروس	With the virus	107.719
12.	خليك	Stay	102.227
13.	عاجل	Urgent	101.479
14.	البيت	The house	94.713
15.	حالات	Cases	90.678
16.	الكورونا	The corona	89.572
17.	جديدة	New	85.742
18.	وفاة	Death	82.464
19.	كلنا	Everyone	81.042
20.	بالبيت	At home	78.395

Individual expressions of loneliness during the pandemic exhibited temporal expansion over four months. Despair in the face of uncertainty was most pronounced in the initial two months when information about the pandemic was scarce, and awareness of the escalating loss of lives was heightened. A conspicuous rise in COVID-19-related

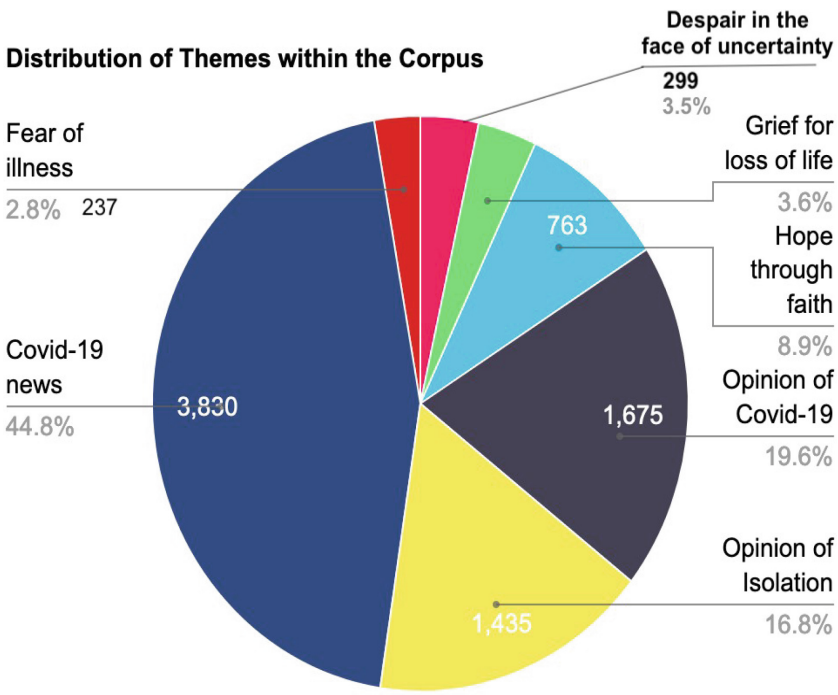

Fig. 1. Distribution of themes within the corpus according to the 7 themes.

news underscored the enduring thematic prevalence of pandemic-related content in public discourse. COVID-19-related news consistently held the highest thematic prevalence. Opinions on COVID-19 and isolation surged throughout the dataset, a trend attributed to lockdown restrictions and the escalating coverage of COVID-19 news. Fear of illness, albeit less prevalent, manifested in individuals who expressed concerns about contracting COVID-19, the potential loss of loved ones, the apprehension of an unending pandemic, and the fear of life never returning to pre-pandemic normalcy. Hope through faith emerged as a resilient theme, encapsulated in tweets that sought strength through religious prayers, extended prayers for isolated individuals or those who succumbed to COVID-19, and expressed hopefulness through prayer for a brighter post-pandemic future.

3.8 Inter Annotator Agreement (IAA)

Two annotators with graduate backgrounds in psychology and over 12 years of training in counseling were tasked to annotate 200 tweets each, categorizing them according to the themes. To assess the reliability of their agreement, Cohen's Kappa was calculated instead of merely relying on the average agreement percentage. The Kappa score, which adjusts for chance agreement, provides a more nuanced understanding of the annotators' reliability.

The calculated Cohen's Kappa score was 0.75, indicating substantial agreement between the annotators, considering that a Kappa value between 0.61 and 0.80 is typically interpreted as substantial agreement. This is a more robust measure than the average agreement score of 82% we obtained initially, as it adjusts for the possibility of chance agreement.

Upon a theme-wise breakdown, the Kappa scores varied, reflecting differing levels of agreement across themes. For instance, the agreement on tweets related to COVID-19 news and hope through faith was particularly high, with Kappa scores suggesting substantial to almost perfect agreement. Conversely, themes like despair in the face of uncertainty and opinions on isolation showed lower Kappa scores, indicating moderate agreement. This analysis via Cohen's Kappa provided a deeper insights into the reliability of the annotations, revealing strengths and areas for improvement in the annotation process.

4 Results

4.1 Corpus Analysis

Or corpus analysis through AntConc demonstrated multiple components about the origin of mental health struggles and loneliness during the initial emergence of COVID-19. Our study employed advanced linguistic analysis techniques. Users expressed diverse forms of loneliness on Twitter, employing various keywords and synonyms. Concordance analysis uncovered 18,241 occurrences of loneliness-related words and enriched our thematic understanding. Emotions were conveyed through themes, and COVID-19 news and opinions were prominently featured. Synonyms of loneliness and the multiple tools available on AntConc led us to direct expressions of mental health struggles and expressions of loneliness. Loneliness was eminently expressed from the start of the pandemic.

The expressions of loneliness manifested directly and diversely in Arabic social media discourse during the initial months of the COVID-19 pandemic. The advanced search option in AntConc directed the study to the Concordance tab, offering the frequency of each word and aiding in finding relevant tweets. The combined frequency of loneliness-related words in Table 2 was 18,299. Selecting the word لوحدي(alone) led to tweets associated with the term. A sample of results included tweets expressing post-COVID plans and reflections on the impact of the pandemic. The word الوحيد(lonely and alone) had a frequency of 2968, with associated tweets illustrating feelings of isolation and unique experiences. Table 3 presented concordance search results for each loneliness-related word and translated tweet examples, enhancing the understanding of their contextual usage.

Corona (كورونا) emerged as the most frequent collocate for words like لوحدي (alone), وحيد (lonely), وحيدة (lonely and loner), واحد (lonely), الوحيدة (lonely), and الوحيد (lonely and alone). Those for الوحيد (alone or lonely, masculine) and الوحيدة (alone or lonely, feminine) highlighted the association between the words and the pandemic, revealing expressions of concern, indifference, and personal experiences related to COVID-19. The top 20 keywords listed in Table 3 highlighted clusters related to COVID-19 and public health (Table 3). كورونا (corona) emerged as the top keyword, reflecting the COVID-19 focus. A separate cluster included words related to public health, and the hashtag #خليك - بالبيت (#stayathome) emphasized staying at home due to lockdown or isolation. The analysis demonstrated the association between keywords and loneliness expression.

Based on AntConc's advanced search, all words were loaded simultaneously to get an overall idea (17367 hits). Each word was searched separately. الوحيدة, defined as a variation of lonely, could also mean "the only thing." KWIC search helped filter 905 hits, revealing instances where words were used without a context of loneliness. This approach differentiated loneliness-related tweets from those unrelated, providing a nuanced analysis. Individuals employed expressions related to loneliness and its synonyms to articulate their emotional experiences during the early stages of lockdown measures in different ways. Our study was focused on the origin of loneliness during the pandemic. Within the first four months of the lockdown period, we found the mental health expression within the dataset consisted of humor, sarcasm, sorrow, grief, and longing.

4.2 Thematic Analysis

Our study identified seven overarching themes, each contributing to the multifaceted emotional landscape during this period. We inferred from the identified themes that COVID-19-related news consistently dominated discussions, indicating the sustained impact of the pandemic on public awareness. While grief for the loss of life was not the primary focus, the staggering global mortality reported by the WHO emphasized the profound impact. Fear of illness, though less prevalent, manifested in concerns about contracting COVID-19, the potential loss of loved ones, and the fear of a prolonged pandemic. Hope through faith emerged as a resilient theme, expressed in tweets seeking strength through religious prayers and anticipating a brighter post-pandemic future. The temporal trends illuminated complex emotional responses during the initial pandemic phase and highlighted social media's adaptability in capturing diverse human emotions amid unprecedented global events.

The Inter Annotator Agreement revealed the potential for varied interpretations among annotators, but a consensus was apparent, particularly in understanding opinions on COVID-19 and COVID-19-related news posts. Through our analyses, expressions of loneliness were prevalent from the onset of the pandemic, persisting throughout the initial lockdown months. Different Arabic variations of the term loneliness were employed, resulting in various tweets and expressions.

5 Discussion

Studies have examined mental health challenges and loneliness amidst crises and pandemics in a broader light. Our study conducted a more in-depth analysis by delving deeply into understanding the articulation of loneliness and exploring how individuals express their feelings during isolation. We derived insightful inferences from thematic elements associated with COVID-19-related loneliness within the context of the Arabic-speaking population.

The increased usage of social media during a global health crisis provided useful resources for emotional, informational, and peer support [16]. The concordance feature is pivotal in facilitating thematic analysis, providing access to tweet segments for examination based on keywords and keyness levels. Our research sought to address a gap in the existing corpus analysis by exploring loneliness in Arabic, a language for which

no such analysis was available online. Through this investigation, we gained valuable insight into the experiences of individuals in the Arabic-speaking community during the pandemic, identifying areas for further research and potential avenues for support and intervention.

There is a correlation between the duration of time spent in social isolation and the onset of loneliness. Spending more than 75% of one's time alone was associated with significantly higher loneliness scores [17]. Our study discerned that manifestations of loneliness in Arabic social media discourse during the initial months of the COVID-19 pandemic were evident from the commencement of lockdown measures, demonstrating a unique and diverse array of expressions. Users exhibited a lack of conformity to a singular form of expression and employed various keywords and multiple identified synonyms associated with loneliness to articulate their thoughts and emotions during this critical lockdown period.

Our study harnessed AntConc's concordance tab to examine the frequency of loneliness-related words in the target corpus. A total of 18,241 occurrences were complemented by examples showcasing tweets associated with specific words and offering insights into the diverse contexts surrounding expressions of loneliness. Concordance search results for each word contributed valuable insights into the prevalence and nuances of loneliness expressions within the dataset. This analytical feature facilitated the identification of pertinent tweets and enriched our understanding of the multifaceted contexts in which loneliness was articulated on social media during the specified timeframe.

Users employed expressions related to loneliness and its synonyms to convey emotional experiences during the early stages of lockdown measures. Expressions spanned a spectrum from humor and fear to sarcasm and sorrow. The lack of expression was observed through retweeting or sharing news updates. Themes played a pivotal role, with the perspective transforming from mere COVID-19 news to opinions when added to a news update or a retweet. The demonstration of user expression through the KWIC search highlights the prevalence of emotional expression by utilizing diverse AntConc features.

We conducted a thematic analysis of themes to identify, analyze, and interpret patterns of meaning [18]. Our analysis revealed distinct themes that exhibited varying prevalence within 8,000 tweets. A predominant theme permeating the corpus is COVID-19 news, indicating the pervasive influence of pandemic-related information. The theme of opinions on COVID-19 emerged prominently, and Twitter serves as a platform to articulate users' thoughts and expressions regarding the emergence and comprehension of the virus.

The process of theme identification offers valuable insights into pertinent patterns within the data, especially during the lockdown period of COVID-19, demonstrating that while there is a prevalence of mental health struggles expressed during this early stage of the pandemic, it was done through different themes. This analytical approach deepens our comprehension by elucidating that individuals, while expressing thoughts, emotions, or news related to COVID-19, imply that users' experiences of loneliness were intricately influenced by these overarching themes. Users whose themes were characterized as despair in the face of uncertainty (3.5%), grief for the loss of life (3.6%), or fear of illness (2.8%) exemplified a tangible connection between prevailing themes and the

articulation of loneliness in their online expressions. This refined perspective sheds light on the complex interplay between broader thematic elements and the manifestation of loneliness within the unique context of the COVID-19 lockdown.

6 Conclusion

Our comprehensive analysis utilizing AntConc has provided in-depth insights into the evolution of mental health challenges and loneliness amidst the early stages of the COVID-19 pandemic. This study not only uncovers the nuanced ways in which individuals in the Arabic-speaking online community expressed and experienced loneliness during these trying times but also enriches the broader scientific dialogue concerning mental health, the dynamics of social media, and the emotional landscape shaped by unprecedented global challenges.

The findings of our study underscore the significant role that language and cultural context play in the articulation and experience of loneliness and mental health struggles. By applying our analytical methodology across different linguistic and cultural landscapes, researchers can uncover the universal and unique dimensions of these experiences, enhancing our global understanding of the pandemic's psychological repercussions.

Moreover, this research contributes to a more empathetic understanding of the human condition in times of crisis, offering valuable perspectives for policymakers, mental health professionals, and community leaders in addressing the mental health needs of diverse populations. Our study advocates for a more interconnected and culturally informed approach to mental health research, emphasizing the importance of considering the linguistic and cultural nuances that shape human experiences and resilience in the face of global adversities.

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Ethical Statement

In conducting this study, we meticulously adhered to ethical standards pertinent to research involving social media data. Given that our analysis solely utilized publicly available social media content, direct interaction with individuals was not a component of our methodology. Nonetheless, we recognize the sensitivity and personal nature of the content analyzed, which reflects real individuals' experiences and sentiments during the COVID-19 pandemic.

To uphold ethical integrity, we ensured the anonymization of all data, carefully removing any potentially identifying information to preserve the privacy and confidentiality of the individuals behind the posts. Our approach was guided by principles of respect for the dignity and anonymity of these individuals, emphasizing the non-intrusive nature of our data collection and analysis.

We also acknowledge our ethical duty to accurately and responsibly interpret the social media data, avoiding misrepresentation or overgeneralization of the findings. Our objective is to contribute to the broader understanding of mental health trends and sentiments during the pandemic, providing insights that are both respectful and reflective of the actual expressions and experiences shared online.

Through this research, we aim to foster a more nuanced understanding of public sentiment and mental health discourse during the COVID-19 pandemic, contributing to the field with an awareness of the ethical implications of working with user-generated content. Our commitment to ethical rigor ensures that our findings not only advance academic knowledge but also respect the inherent dignity and privacy of individuals whose digital footprints inform our analysis.

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